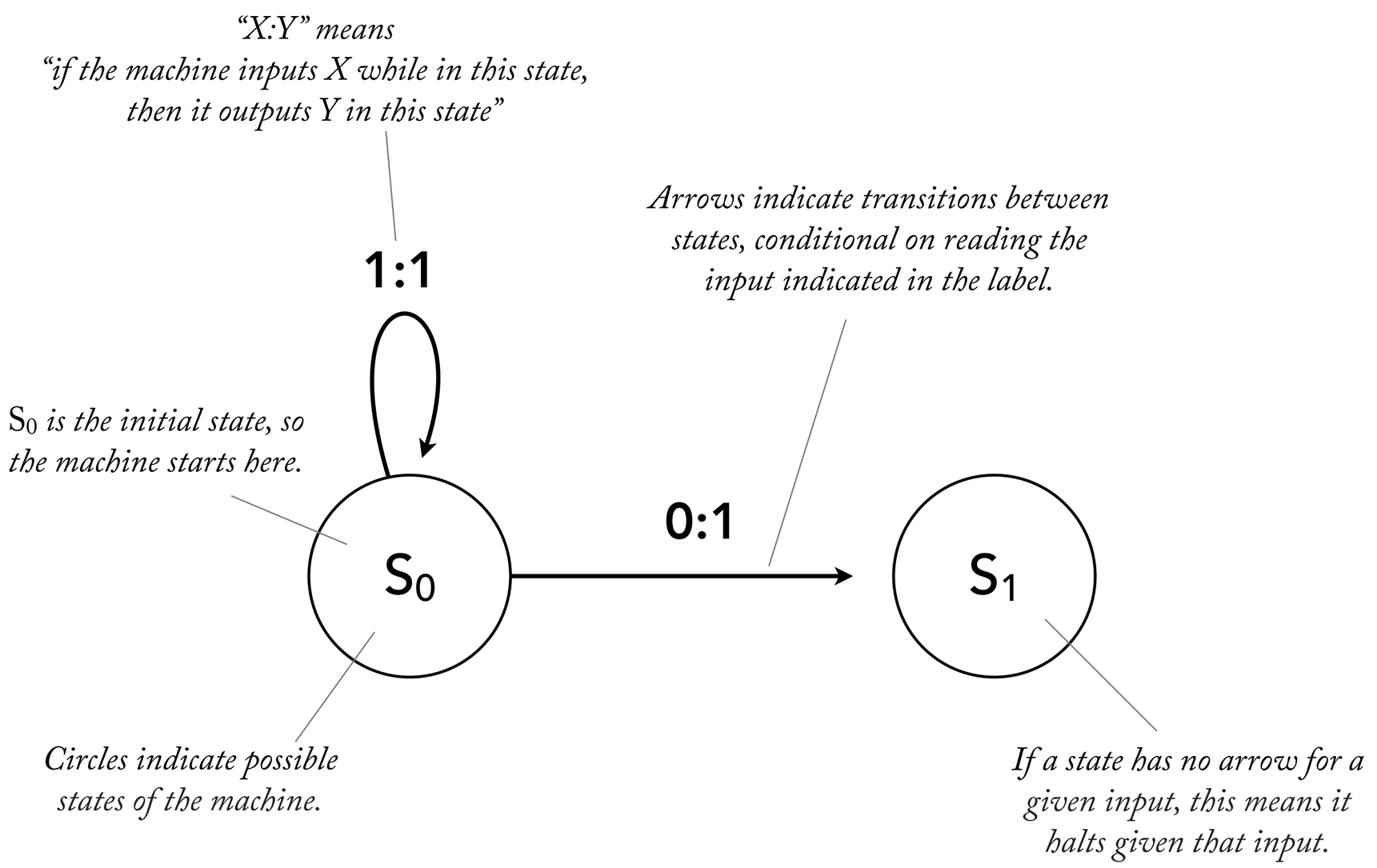
# 24. Finite State Machines

## Reading FSM Diagrams

You can get a feel for how FSM diagrams work by studying the following simple example:




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This diagram represents the following rule:

In state $S_0$,
if input = 1, then output = 1, and return to $S_0$;
if input = 0, then output = 1, and move to $S_1$.

In state $S_1$, halt on any input.

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Computation for FSMs is defined in the same way as it is for SCs. Local inputs and outputs are processed one bit at a time, and the global input and output corresponds to the sequences of inputs and outputs which is reached when all further input and output bits will be 0s.

## Constructing FSM Diagrams

FSM diagrams can be constructed from the following elements.

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**Nodes:** circles associated with names.

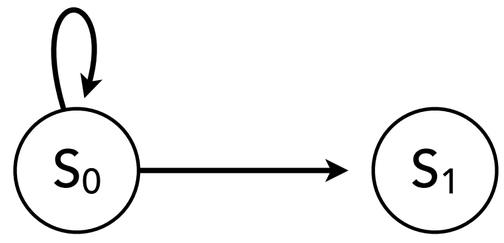
- Each node represents a state.
- The initial state is always named "$S_0$".




---

**Links:** one-directional arrows connecting circles to circles.

- There are at most two lines leaving each circle.*
- There may be zero lines leaving a circle.
- Circles can be connected to themselves.
- The lines can cross.

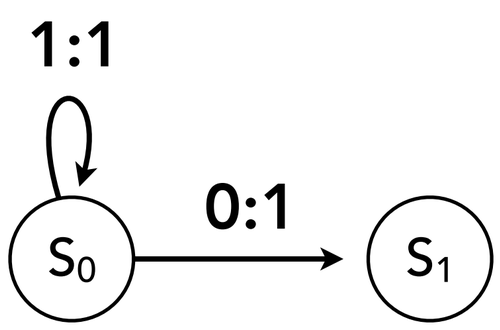



*More precisely, there are as many lines leaving each circle as possible inputs. If the machine can read *n* symbols, there will be up to *n* links for each node.

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**Labels:** each line is labeled with an expression of the form "*input* : *output*".

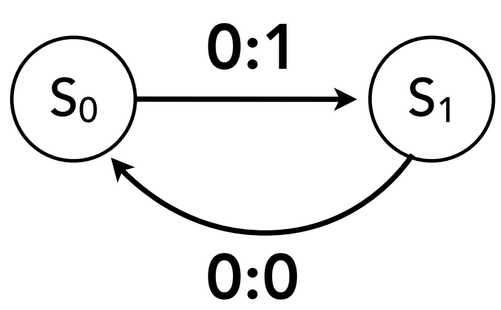
- Possible inputs: 1, 0
- Possible outputs: 1, 0
- **Note!** For each circle, there can be at most one arrow labeled with input 1, and one arrow labeled with input 0.
- A common variation writes the output where we've put the state name, and labels the line only with the input.




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**Loops:** any kind of loop is acceptable:

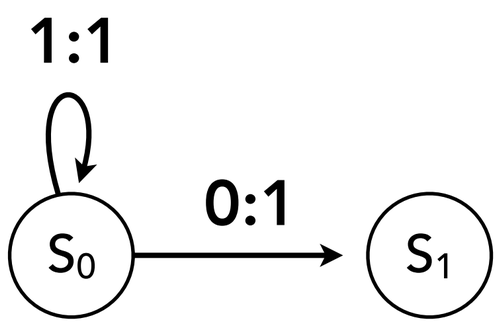
- Loops from a state to itself.
- Loops among a group of states.



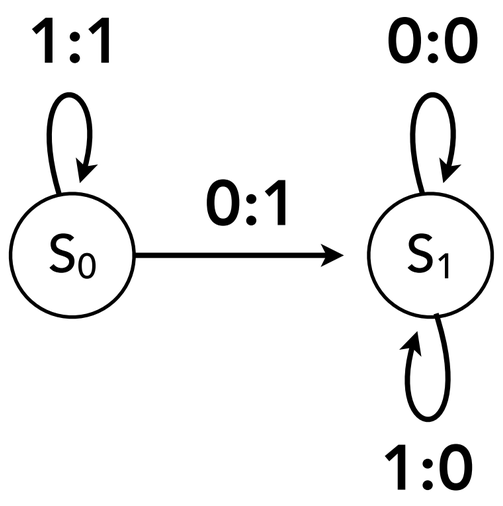

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**Halting:**

- A machine halts in a given state, given an input, if there are no arrows leaving that state for that input.
- If there are *no* lines leaving a state, the machine will halt in that state no matter what.
- When computing a function, an FSM should never halt. Instead, it should end computation by outputting 0s indefinitely.




This machine halts.

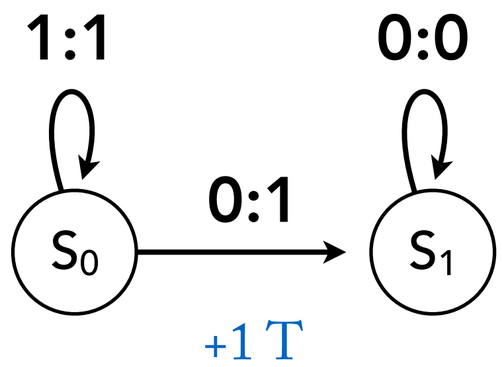



This machine never halts.

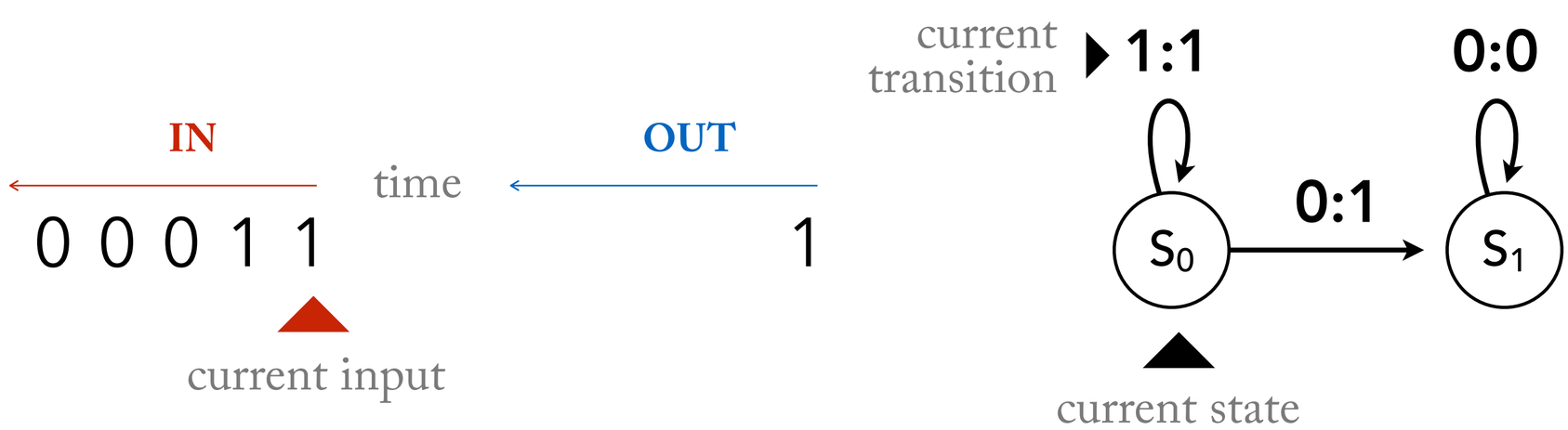
## Example: +1 T

This FSM at right computes successor in tally.

The timeline below shows the step-by-step progression of the machine for a sample input.




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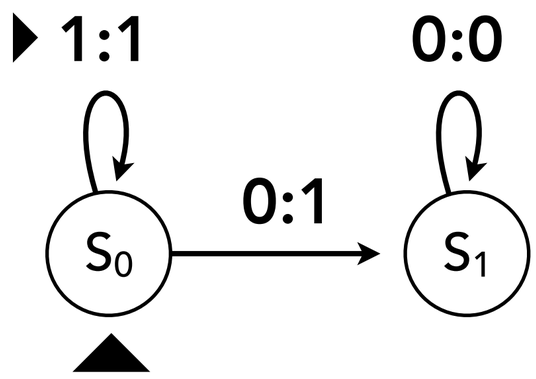

---

IN

0 0 0 1 1

OUT

1 1



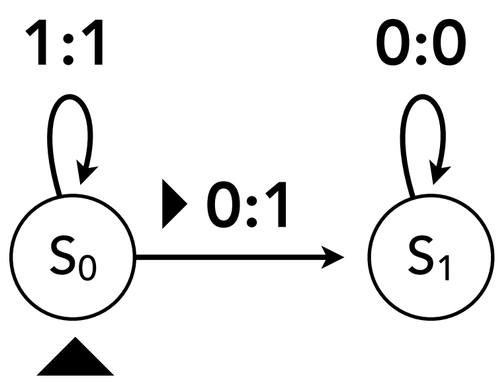

---

IN

0 0 0 1 1

OUT

1 1 1



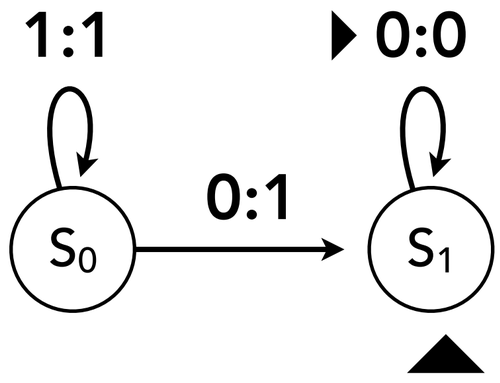

---

IN

0 0 0 1 1

OUT

0 1 1 1



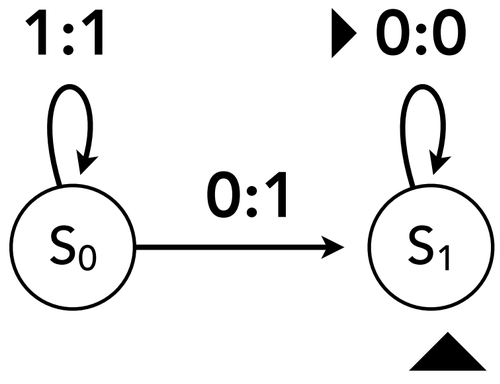

---

IN

0 0 0 1 1

OUT

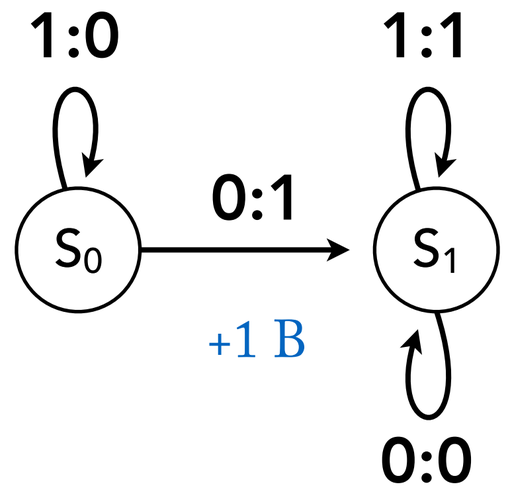
0 0 1 1 1



## Example: +1 B

This FSM at right computes successor in binary.

The timeline below shows the step-by-step progression of the machine for a sample input.



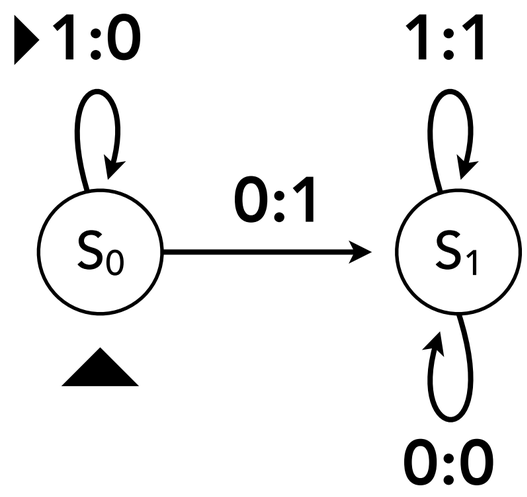

---

IN

0 0 1 1

OUT

0



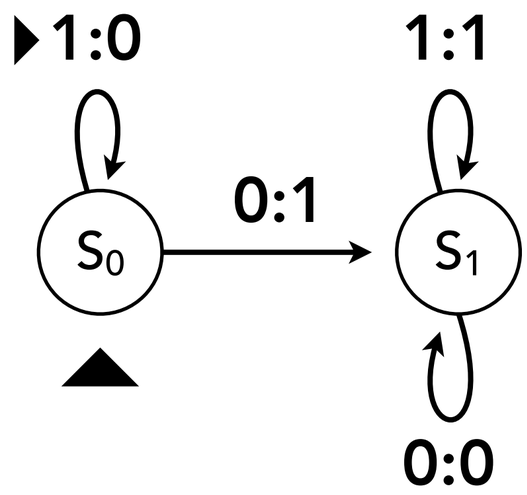

---

IN

0 0 1 1

OUT

0 0



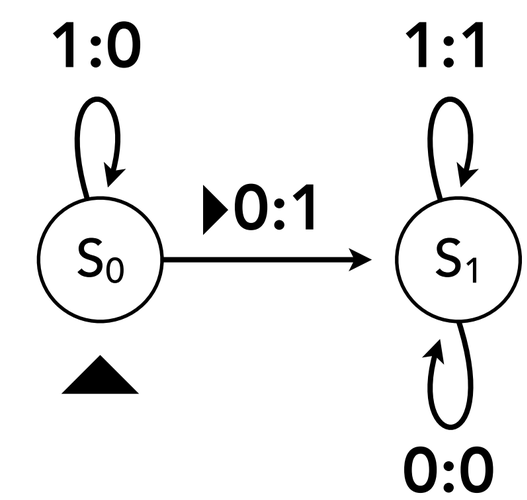

---

IN

0 0 1 1

OUT

1 0 0




---

IN

0 0 1 1

OUT

0 1 0 0

